Fast Line Recognition on the CPU

1. Introduction

Optical character recognition turns the image of a page into text that a computer can search, index and edit. A fast recognizer reads a line at a time and needs only a small network.

The detector finds every line of text on the page and returns its box. Each box is cropped, scaled to a fixed height and read by the recognizer, one character per column of features.

2. Method

Reading order decides which line comes next. In a two-column article the reader goes down the left column first and then continues at the top of the right column.

1. Detect the lines of text.
2. Recognize each line.
3. Order the lines and blocks.
4. Write markdown.

Layout classes group lines into blocks: headings, paragraphs, lists, tables and captions. Page headers and footers are left out of the markdown unless they are asked for.

3. Results

Table 1 compares the throughput of three engines on the same machine.

Engine	Pages/s	Score
Indra	49.4	62.1
Tesseract	2.9	19.3
PaddleOCR	4.3	35.8

The fastest engine reads a page in about a quarter of a second on four cores.